

Algorithm 1: Inventory Management System

Exercise 1: Inventory Management System

Scenario:

You are developing an inventory management system for a warehouse. Efficient data storage and retrieval are crucial.

Steps:

Understand the Problem:

Explain why data structures and algorithms are essential in handling large inventories.

Discuss the types of data structures suitable for this problem.

Setup:

Create a new project for the inventory management system.

Implementation:

Define a class Product with attributes like productId, productName, quantity, and price.

Choose an appropriate data structure to store the products (e.g., ArrayList, HashMap).

Implement methods to add, update, and delete products from the inventory.

Analysis:

Analyze the time complexity of each operation (add, update, delete) in your chosen data structure.

Discuss how you can optimize these operations.